

SEAT No. _____

No. of Printed Pages: 02

[108]

SARDAR PATEL UNIVERSITY
B.C.A (SEMESTER-VI) EXAMINATION 2024
US04CBCA25 : Data Structure - II



Date : 30/03/2024 (Saturday)

Time : 02:00 p.m. to 05:00 p.m.

Marks: 70

Q. 1 Multiple Choice Questions**[10]**

- 1 A binary tree has at most _____ child.
a. Two. b. Three. c. One. d. None of these.
- 2 _____ is a set of disjoint trees.
a. Path. b. Directed tree. c. Forest. d. Height.
- 3 Leaf node is known as _____ node.
a. Terminal. b. Parent. c. Root. d. Non-terminal
- 4 A linked list is _____ type of data structure.
a. Linear b. Non-Linear c. Both (A) and (B) d. None of these
- 5 In linked list each node contain minimum of two fields. One field is data field to store the data and the second field is _____.
a. Pointer to character b. Pointer to node
c. Pointer to integer d. Node
- 6 Each node in singly linked list has _____ fields.
a. 2 b. 3 c. 1 d. 4
- 7 _____ is the process to arrange the data in ascending or descending order.
a. Sorting b. Inserting c. Searching d. Updating
- 8 Which design technique is used for Quick Sort...?
a. Divide and Conqueror b. Dynamic Programming
c. Greedy d. Backtracking
- 9 A _____ technique resolves conflicts that arise when more than one record key is mapped to the same table location.
a. Hashing b. Blocking c. Collision-resolution d. None of these
- 10 The number of records in a bucket is called the _____.
a. Bucket capacity b. Table capacity c. File Capacity d. None of these

Q. 2 Short Question (Any TEN)**[20]**

- 1 What is Binary Tree? Also give its example.
- 2 List out Applications of Tree.
- 3 Define the Terms: 1. Root 2. Node
- 4 Explain linked list in brief.
- 5 Explain circular linked list in brief.
- 6 Lists insert operations that can be performed on singly linked list.
- 7 Define sorting. List out the names of sorting techniques
- 8 Advantages and Disadvantages of Bubble sort & Merge Sort
- 9 List out Categories of Sorting. Define any one of them.
- 10 Define: Hashing function, Collision resolution technique
- 11 List types of Chaining
- 12 List the types of overflow are used in indexed sequential file organization.

Q. 3

[A] Explain the insertion of a node in a binary with an example. [05]

[B] Draw a binary tree for the expression $A*B-(C+D)*(P/Q)$. [05]

OR

Q.3

[A] Explain deletion of a node from binary tree with example. [05]

[B] Draw a binary tree for following equation. [05]

1. $(A+B*C)/(A+B)*C$

2. $A+(B-C)*(E*F)/(G/J)$

Q. 4

[A] Write an algorithm to insert node at front position in singly linked list. [05]

[B] Write an algorithm to delete node at any position in singly linked list. [05]

OR

Q. 4

[A] Write an algorithm to delete node at beginning in singly linked list. [05]

[B] Write an algorithm to insert node at end position in singly linked list. [05]

Q. 5

[A] Write down the algorithm of Merge sort. [05]

[B] Write down the algorithm of Binary Search. [05]

OR

Q. 5

[A] Write down the algorithm of Linear Search. [05]

[B] Write down the algorithm of Quick sort [05]

Q. 6 List and explain popular hashing function with example. [10]

OR

Q. 6 Write a detail note on structure and processing of Direct files. [10]

